

Predicting Academic Burnout in Secondary and Undergraduate Students Using Explainable Machine Learning: Roles of Academic Stress, Psychological Distress, Self-Efficacy, and Social Support

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ABSTRACT

Academic burnout has emerged as a growing concern among secondary school and undergraduate students, particularly in competitive, achievement-oriented educational environments marked by high performance expectations and examination pressure. Conceptualised as a multidimensional syndrome involving emotional exhaustion, cynicism, and diminished academic efficacy, burnout is consistently linked to reduced academic performance, lower motivation, and impaired psychological well-being. While prior research has established clear relationships between academic stress, anxiety, self-efficacy, and burnout, existing studies rely predominantly on explanatory statistical models with limited predictive capability for identifying at-risk students before consequences become severe.

The present study developed and evaluated machine learning models to predict academic burnout among secondary school (Grades 9–12) and first-year undergraduate students in India. Survey data from 321 students measured academic stress, psychological distress, self-efficacy, social support, sleep patterns, and screen time. Four classifiers — Logistic Regression, SVM, Random Forest, and XGBoost — were trained and compared using five-fold stratified cross-validation, with performance evaluated on accuracy, precision, recall, F1-score, and ROC-AUC. SHAP analysis was used to ensure interpretability and quantify each predictor's contribution.

Random Forest achieved the highest performance (Accuracy = 97.5%, F1 = 0.977, ROC-AUC = 0.998), with XGBoost performing comparably (Accuracy = 96.3%, ROC-AUC = 0.994). SHAP identified psychological distress as the dominant predictor (Mean |SHAP| = 0.231), followed by social support (0.163) and self-efficacy (0.110) as protective factors, with academic stress showing the weakest unique contribution (0.056). Regression confirmed that distress ($B = 0.356$, $p < .001$), self-efficacy ($B = -0.308$, $p < .001$), and social support ($B = -0.314$, $p < .001$) each independently predicted burnout, explaining 58.3% of variance ($R^2 = .583$). These findings highlight the value of combining machine learning with explainable AI for early identification of at-risk students.

KEYWORDS

academic burnout, secondary school students, undergraduate students, psychological distress, self-efficacy, social support, machine learning, SHAP, explainable artificial intelligence, India.

INTRODUCTION

The academic experiences of secondary school and undergraduate students have undergone considerable transformation in recent decades. Expanding competition for university admissions, high-stakes examinations, and intensifying achievement expectations have produced educational environments characterised by sustained performance pressure. While academic aspiration serves important developmental functions, prolonged exposure to these pressures exacts a substantial psychological toll: students across diverse educational systems report elevated academic stress, anxiety, emotional exhaustion, and disengagement — a constellation of symptoms increasingly recognised as academic burnout [1–3].

The consequences of academic burnout extend well beyond immediate performance deficits, encompassing reduced motivation, poorer examination outcomes, increased dropout intentions, and heightened vulnerability to depression, anxiety, and chronic stress [2]. These wide-ranging consequences render early identification of at-risk students not merely an educational concern, but a public health imperative.

Despite substantial theoretical and empirical progress in understanding burnout's psychological correlates — including academic stress, psychological distress, self-efficacy, and social support — the majority of existing research relies on explanatory statistical methods designed to elucidate relationships among variables rather than predict individual-level outcomes. This predictive gap has real consequences: interventions are most effective when targeted at genuinely at-risk individuals before consequences materialise, yet without reliable predictive tools, educators are largely confined to reactive rather than proactive responses.

Recent advances in machine learning offer a promising avenue for addressing this limitation, given their capacity to model complex, nonlinear relationships and generate individual-level predictions from multidimensional data. However, comparatively few studies have applied machine learning specifically to

academic burnout prediction, and those that have done so have focused predominantly on university populations in Western contexts — leaving secondary school and first-year undergraduate populations in non-Western, high-pressure educational contexts such as India underexplored. A further limitation of many high-performing algorithms is their "black box" nature; explainable AI techniques such as SHAP (SHapley Additive exPlanations) [11] address this by quantifying each predictor's contribution to individual outcomes, enabling findings to be situated within established theoretical frameworks.

The present study addresses these gaps by developing and evaluating multiple machine learning models to predict academic burnout among secondary school and undergraduate students in India, using SHAP analysis to ensure interpretability. Survey data from 321 students were used to train four classifiers — Logistic Regression, Support Vector Machine, Random Forest, and XGBoost — evaluated using five-fold stratified cross-validation.

This study makes four principal contributions. First, it extends machine learning burnout prediction to a secondary school and early undergraduate sample in an Indian educational context, a population underrepresented in the literature. Second, it integrates SHAP-based explainable AI alongside predictive modelling. Third, it examines a theoretically grounded set of psychosocial predictors spanning both risk and protective factors. Fourth, it provides preliminary evidence on the relative predictive importance of psychological distress, social support, self-efficacy, and academic stress, with direct implications for targeted early-intervention design.

The paper proceeds as follows. Section 2 reviews the relevant literature on burnout theory, psychological distress, self-efficacy, social support, and machine learning in educational contexts. Section 3 describes the methodology. Section 4 presents the results. Section 5 discusses findings in relation to prior research, with Section 5.5 addressing limitations. The Conclusion offers practical implications for educators, counsellors, and institutions.

LITERATURE REVIEW

2.1 Conceptualising Burnout: From Occupational to Academic Contexts

The systematic study of burnout originated within occupational psychology. Maslach and Jackson [1] provided the first rigorous operationalisation of burnout as a three-dimensional syndrome: emotional exhaustion (a pervasive sense of being emotionally depleted), depersonalisation (later reconceptualised as cynicism in academic contexts, involving detachment and disengagement), and reduced personal accomplishment (diminished self-perceptions of competence and achievement). This three-dimensional structure has demonstrated considerable cross-cultural replication and remains foundational to contemporary burnout research.

Maslach, Schaufeli, and Leiter [6] later situated this framework within a broader model of chronic stress and coping resource depletion, emphasising that burnout develops progressively from sustained mismatches between environmental demands and available coping resources — a perspective with particular relevance to educational contexts, where cumulative academic pressure may gradually exceed students' adaptive capacity.

The translation of burnout theory to educational settings was advanced through student-specific instruments. Schaufeli et al. [5] adapted the Maslach Burnout Inventory into the Maslach Burnout Inventory–Student Survey (MBI-SS), conceptualising student burnout across exhaustion, cynicism, and academic efficacy. Salmela-Aro et al. [13] subsequently developed the more concise School Burnout Inventory (SBI) for secondary school populations. Both instruments have strong psychometric properties across international samples, and the present study draws on items reflecting all three dimensions in its composite burnout measure.

2.2 Academic Burnout Among Secondary School and Undergraduate Students

Bask and Salmela-Aro [4] found in a longitudinal study of Finnish adolescents that prolonged academic stress produced progressively increasing exhaustion and disengagement across the academic year — a cumulative pattern consistent with Maslach et al.'s [6] theoretical account. Walburg [2], reviewing burnout across national contexts, identified persistent academic pressure, high parental expectations, and limited institutional support as key drivers, with particular relevance to the Indian context given its competitive entrance-examination culture.

Among undergraduates, burnout has been associated with increased dropout intentions and compromised mental health, with the secondary-to-university transition representing a period of heightened vulnerability. Burnout dimensions appear to manifest somewhat differently by level: secondary students often report exhaustion from structured examination demands, while undergraduates exhibit more pronounced cynicism from misalignment between educational experience and personal goals.

Gao et al. [3] provided mechanistic evidence via a moderated mediation framework, showing that academic stress predicted burnout both directly and indirectly through academic anxiety, with self-efficacy moderating the stress–anxiety relationship. This model directly informs the theoretical positioning of the present study.

2.3 Academic Anxiety, Psychological Distress, and the Stress–Burnout Pathway

Academic anxiety — apprehension and worry regarding performance and evaluation — is one of the most consistently identified mechanisms linking academic stress to burnout, impairing attentional focus and academic confidence over time. Pekrun's [12] Control-Value Theory frames this process theoretically: academic emotions arise from students' appraisals of perceived control and the value assigned to academic tasks, with high stakes combined with low perceived control (common under examination pressure) producing achievement-related anxiety that depletes the emotional reserves needed for sustained engagement.

Psychological distress, though related, is broader — encompassing general difficulty with emotional regulation and coping under pressure from sources beyond academic demands (e.g., family or interpersonal stress). The present study assesses it as a construct distinct from academic stress, allowing the unique contribution of this broader emotional burden to be examined separately.

2.4 Self-Efficacy as a Protective Factor Against Burnout

Bandura's [7,8] social cognitive theory identifies self-efficacy — belief in one's capability to execute actions required for desired outcomes — as central to resilience under stress, shaping whether demanding situations are perceived as challenges or threats and determining persistence in the face of setbacks. Academic self-efficacy specifically has been consistently identified as protective against burnout: students with stronger efficacy beliefs show greater resilience under examination pressure and reduced emotional depletion. Gao et al. [3] found direct support for this, with self-efficacy moderating the stress–burnout relationship such that high-efficacy students showed substantially attenuated burnout under stress.

The present study operationalises self-efficacy through students' confidence in handling difficult assignments and achieving academic goals, generating the prediction that higher self-efficacy will emerge as a protective predictor in both regression and machine learning analyses.

2.5 Social Support and Burnout Risk

Perceived social support — the belief that emotional and practical assistance is available from family, peers, and educators — buffers the effects of chronic stress by reducing perceived threat, easing objective workload burden, and fostering belonging that sustains engagement. This is particularly salient for secondary and early undergraduate students, for whom family and peer networks are primary sources of support. The present study assesses social support through perceived family support during difficult periods and awareness of available help resources.

METHODS AND MATERIALS

3.1 Participants

Participants were 321 students recruited through a voluntary, anonymous online survey distributed via Google Forms across school networks, social media platforms, and peer referral channels in India. The sample comprised students currently enrolled in Grades 9–12 (secondary school) and first-year undergraduate programmes, representing the two educational levels at which academic pressure and burnout risk are considered particularly pronounced. No exclusion criteria beyond current enrolment at these educational levels were applied. The sample constitutes a convenience sample; accordingly, findings should be interpreted with appropriate caution regarding population representativeness and external generalisability.

Prior to completing the survey, all participants were presented with an informed consent statement describing the study's academic purpose, the entirely voluntary nature of participation, the anonymity and confidentiality of all responses, and participants' unrestricted right to withdraw at any stage without consequence. No personally identifiable information was collected at any point during the study. Given the inclusion of items related to psychological distress and emotional well-being, the consent statement noted that the study was designed for research purposes only and that responses would not be shared with teachers, parents, or any institutional body. Participants were also reminded of the availability of general mental health resources should they experience any distress during completion.

This study did not undergo formal review by an Institutional Review Board (IRB), as it was conducted as independent student research without formal university or institutional affiliation. Ethical safeguards were nonetheless followed throughout, including informed consent, voluntary participation, anonymity of responses, and the right to withdraw at any stage, as described above.

3.2 Measures

3.2.1 Academic Stress

Academic stress was assessed using five items reflecting core dimensions of academic stress identified in the literature: perceived frequency of feeling overwhelmed by schoolwork, perceived pressure to achieve high grades, worry about academic performance, worry about upcoming tests and

examinations, and difficulty balancing schoolwork with other responsibilities. Items were rated on a five-point Likert-type scale ranging from 1 (Not at all / Strongly Disagree) to 5 (Very Often / Strongly Agree). A composite score was computed as the unweighted mean of all five items, with higher scores indicating greater academic stress. Internal consistency was good (Cronbach's $\alpha = .823$, 95% CI [.790, .852]).

3.2.2 Psychological Distress

Psychological distress was assessed using three items measuring the self-reported frequency of feeling emotionally overwhelmed, struggling to cope with stress, and experiencing persistent worry over the preceding month. Items were rated on the same five-point scale. A composite distress score was computed as the mean of the three items. Internal consistency was good (Cronbach's $\alpha = .827$, 95% CI [.791, .857]).

3.2.3 Academic Burnout

Academic burnout was operationalised as a composite score reflecting three theoretically defined dimensions, adapted from the Maslach Burnout Inventory–Student Survey [5]. The Exhaustion subscale comprised three items assessing emotional and physical depletion arising from academic demands (e.g., feeling exhausted at the end of a typical school day, feeling emotionally drained by studies, feeling too tired to complete academic tasks). The Cynicism subscale comprised two items assessing disinterest in and questioning of the meaningfulness of studies. The Academic Efficacy subscale comprised two items assessing academic confidence and goal attainment, which were reverse-scored prior to inclusion in the composite so that higher values reflected lower efficacy — consistent with the burnout framework in which reduced efficacy is indicative of greater burnout severity. A composite burnout score was computed as the unweighted mean of all seven items (including reverse-coded efficacy items), with higher scores indicating greater burnout. Internal consistency for the composite was questionable (Cronbach's $\alpha = .655$, 95% CI [.595, .710]), likely reflecting the construct's multidimensionality and the limited number of items per subscale; this limitation is discussed in Section 5.5.

3.2.4 Self-Efficacy

Self-efficacy was assessed using two items measuring students' confidence in their ability to handle difficult academic assignments and their belief in their capacity to achieve academic goals, rated on the five-point scale. A composite score was computed as the mean of the two items. Cronbach's alpha was low ($\alpha = .265$), attributable primarily to the structural constraint imposed by two-item scales, for which alpha is inherently suppressed independent of item quality. The inter-item correlation was $r = .26$. These limitations are discussed in Section 5.5.

3.2.5 Social Support

Social support was assessed using two items measuring perceived family support during difficult academic periods and students' awareness of where to seek help when facing academic challenges. A composite score was computed as the mean of both items. Cronbach's alpha was low ($\alpha = .572$), again primarily reflecting the two-item constraint. These limitations are discussed in Section 5.5.

3.2.6 Sleep and Screen Time

Sleep duration was assessed by asking participants to indicate their average nightly sleep from a set of categorical response options (less than 5, 4–5, 5–6, 6–7, 7–8, and more than 8 hours). Screen time was assessed via two items measuring total daily screen time outside of schoolwork and daily social media usage, both using categorical response formats. These variables were included in the survey for descriptive and exploratory purposes; due to their categorical encoding and the methodological decision to restrict the primary machine learning analyses to continuous composite scale scores, they were not incorporated into the main predictive models. Their inclusion in future analyses using appropriate encoding strategies is recommended.

3.3 Procedure

The survey was administered electronically via Google Forms and distributed to potential participants through secondary school networks, undergraduate institution contacts, social media platforms, and peer referral. Data collection proceeded until a sample of 321 usable responses had been

obtained following the removal of duplicate submissions identified through response pattern analysis. No monetary or academic compensation was provided for participation.

Scale scores were computed by averaging the relevant item responses for each construct. For the burnout composite, self-efficacy items were first reverse-scored (by subtracting the original score from 6) prior to averaging. Complete case analysis was applied; participants with missing values on any item contributing to a given scale were excluded from analyses involving that scale. Given the very low overall rate of missing data ($< 1\%$), this approach had a negligible impact on statistical power.

3.4 Machine Learning Pipeline

3.4.1 Target Variable and Binarisation

For machine learning classification, the continuous burnout composite score was dichotomised using a median split. Participants scoring at or above the sample median burnout score ($Mdn = 3.29$) were assigned to the High Burnout class ($n = 181$; 56.4%), and those scoring below the median were assigned to the Low Burnout class ($n = 140$; 43.6%). Median-split binarisation was employed to enable the application of binary classification algorithms and is consistent with approaches adopted in prior burnout classification research. The associated loss of continuous outcome information and implications for performance interpretation are acknowledged as limitations in Section 5.5.

3.4.2 Feature Set

The primary feature set for all machine learning models comprised four continuous composite scale scores: Academic Stress, Psychological Distress, Self-Efficacy, and Social Support. All features were standardized to zero mean and unit variance using z-score normalisation (StandardScaler), applied independently within each cross-validation fold to prevent data leakage from the test set into the training process.

3.4.3 Classification Algorithms

Four machine learning algorithms were trained and evaluated: (1) Logistic Regression — a linear probabilistic classifier providing a theoretically interpretable baseline model with direct correspondence

to regression coefficients; (2) Support Vector Machine (SVM) with a radial basis function (RBF) kernel — a nonlinear classifier that identifies the optimal separating hyperplane in a transformed feature space; (3) Random Forest — a bootstrap-aggregated ensemble of decision trees providing robust performance, native resistance to overfitting through bagging, and inherent feature importance estimates; and (4) XGBoost (Extreme Gradient Boosting) — a sequential gradient boosting algorithm that iteratively constructs an ensemble of weak learners to minimise a differentiable loss function, widely employed for its strong empirical performance across diverse prediction tasks.

3.4.4 Validation and Evaluation

Model performance was evaluated using five-fold stratified cross-validation, in which the dataset was partitioned into five folds while preserving the class distribution of the full sample in each fold. Each model was trained on four folds and evaluated on the remaining fold, with this process repeated five times such that every observation served as part of the test set exactly once. Performance was assessed across five metrics: accuracy (proportion of correctly classified observations), precision (proportion of predicted positive cases that are true positives), recall (proportion of true positive cases correctly identified), F1-score (harmonic mean of precision and recall), and ROC-AUC (area under the receiver operating characteristic curve, reflecting overall discriminative ability across all classification thresholds). Mean performance and standard deviation across the five folds are reported for all models.

3.4.5 SHAP Analysis

Following model evaluation, SHAP (SHapley Additive exPlanations) [11] values were computed for the best-performing model — Random Forest — to quantify the contribution of each predictor to individual burnout classification decisions. The TreeExplainer algorithm, optimised for tree-based ensemble methods, was employed to compute exact SHAP values for all 321 observations. Mean absolute SHAP values were computed for each feature to generate a global feature importance ranking. A SHAP summary plot was additionally generated to visualise both the direction and magnitude of each feature's influence on individual-level burnout predictions, enabling examination of the nature (risk-increasing versus protective) of each predictor's contribution.

All analyses were conducted in Python (version 3.11) using the scikit-learn [18], XGBoost [19], and SHAP [11] libraries. Statistical analyses were conducted using statsmodels [20] and pingouin [21].

RESULTS

4.1 Reliability Analysis

Prior to primary analyses, the internal consistency of each composite scale was evaluated using Cronbach's alpha. Results are presented in Table 1. Academic Stress ($\alpha = .823$) and Psychological Distress ($\alpha = .827$) demonstrated good internal consistency, satisfying the conventional threshold of $\alpha \geq .80$ recommended for research measurement. The Burnout composite yielded a questionable alpha of .655, likely attributable to the construct's inherent multidimensionality and the limited number of items comprising each subscale. Self-Efficacy ($\alpha = .265$) and Social Support ($\alpha = .572$) demonstrated low reliability estimates; however, both are two-item scales, for which Cronbach's alpha is structurally constrained independent of item quality. These limitations are discussed in Section 5.5.

Table 1
Cronbach's Alpha Reliability Estimates for All Composite Scales

Scale	α	95% CI	No. Items	Interpretation
Academic Stress	.823	[.790, .852]	5	Good
Psychological Distress	.827	[.791, .857]	3	Good
Academic Burnout (composite)	.655	[.595, .710]	7	Questionable
Social Support	.572	[.467, .656]	2	Low (2-item) [†]
Self-Efficacy	.265	[.084, .410]	2	Low (2-item) [†]

[†] Two-item scales are subject to structural constraints in alpha estimation; item-level intercorrelations are provided as supplementary reliability evidence in Appendix A.

4.2 Descriptive Statistics

Descriptive statistics for all five composite scale scores are presented in Table 2. All scales were scored on a 1–5 metric. Mean scores clustered around the midpoint of the response scale (range: 3.31–3.67), indicating moderate-to-high levels of academic stress, psychological distress, and burnout in the sample alongside moderate levels of self-efficacy and social support. Distributions were approximately

symmetric with skewness values within acceptable ranges (all $|\text{skew}| < .50$) and mildly platykurtic (negative kurtosis), supporting the application of parametric statistical techniques.

Table 2

Descriptive Statistics for Composite Scale Scores (N = 321)

Scale	M	SD	Min	Max	Mdn	Skew	Kurt
Academic Stress	3.67	0.69	2.60	5.00	3.60	+0.29	-1.10
Academic Burnout	3.31	0.79	1.57	5.00	3.29	-0.34	-0.83
Psychological Distress	3.42	1.08	1.33	5.00	3.33	-0.09	-0.99
Self-Efficacy	3.52	1.04	1.00	5.00	3.50	-0.41	-0.50
Social Support	3.49	0.97	1.50	5.00	3.50	-0.48	-0.75

4.3 Correlational Analysis

Pearson correlations among all five composite scale scores are presented in Table 3. All predictors demonstrated statistically significant bivariate associations with burnout at $p < .001$. Psychological distress showed the strongest positive association with burnout ($r = .563$), followed by academic stress ($r = .441$). Social support ($r = -.470$) and self-efficacy ($r = -.398$) demonstrated significant negative associations, consistent with their theorised protective roles. Notably, academic stress and psychological distress were highly intercorrelated ($r = .683$), indicating substantial shared variance that should be considered when interpreting regression and machine learning results.

Table 3

Pearson Correlation Matrix Among All Scale Scores

	1. Stress	2. Distress	3. Efficacy	4. Support
1. Academic Stress	—			
2. Psychological Distress	.683***	—		

	1. Stress	2. Distress	3. Efficacy	4. Support
3. Self-Efficacy	-.077	-.044	—	
4. Social Support	-.412***	-.289***	-.060	—
5. Academic Burnout	.441***	.563***	-.398***	-.470***

Note. *** $p < .001$.

4.4 Multiple Linear Regression

A multiple linear regression was conducted to examine the independent predictive contributions of academic stress, psychological distress, self-efficacy, and social support to burnout scores (see Table 4). The overall model was statistically significant, $F(4, 316) = 110.41$, $p < .001$, explaining 58.3% of variance in burnout ($R^2 = .583$, Adjusted $R^2 = .578$). Psychological distress ($B = 0.356$, $SE = 0.036$, $t = 9.87$, $p < .001$), self-efficacy ($B = -0.308$, $SE = 0.028$, $t = -11.10$, $p < .001$), and social support ($B = -0.314$, $SE = 0.033$, $t = -9.66$, $p < .001$) each made statistically significant unique contributions. Academic stress did not reach statistical significance as an independent predictor when controlling for the other variables ($B = -0.096$, $SE = 0.060$, $t = -1.62$, $p = .107$), a pattern consistent with the high intercorrelation between stress and distress ($r = .683$) and the likely subsumption of stress variance by the distress measure when both are entered simultaneously.

Table 4

Multiple Linear Regression Predicting Academic Burnout ($N = 321$)

Predictor	B	SE	t	p	Sig.
Intercept	4.620	0.271	17.08	< .001	***
Psychological Distress	+0.356	0.036	9.87	< .001	***
Self-Efficacy	-0.308	0.028	-11.10	< .001	***
Social Support	-0.314	0.033	-9.66	< .001	***
Academic Stress	-0.096	0.060	-1.62	.107	ns

Note. $R^2 = .583$, Adjusted $R^2 = .578$, $F(4, 316) = 110.41$, $p < .001$. *** $p < .001$; ns = non-significant.

4.5 Machine Learning Model Performance

Performance metrics for all four machine learning classifiers, evaluated using five-fold stratified cross-validation, are presented in Table 5. The Random Forest classifier achieved the highest performance across all five metrics (Accuracy = .975, Precision = .989, Recall = .967, F1 = .977, ROC-AUC = .998), demonstrating near-perfect discrimination between high and low burnout classifications. XGBoost performed comparably (Accuracy = .963, ROC-AUC = .994), followed by SVM (Accuracy = .931, ROC-AUC = .959) and Logistic Regression (Accuracy = .879, ROC-AUC = .952). The strong performance of Logistic Regression — the simplest and most interpretable classifier — is noteworthy, suggesting that the relationships between predictors and burnout classification are sufficiently robust to be captured by linear decision boundaries and are consistent with the directional effects identified in the regression analysis.

Table 5

Machine Learning Classification Performance — 5-Fold Stratified Cross-Validation (N = 321)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest*	.975	.989	.967	.977	.998
XGBoost	.963	.970	.967	.967	.994
SVM (RBF kernel)	.931	.940	.939	.939	.959
Logistic Regression	.879	.899	.895	.894	.952

*Note. Values represent means across five cross-validation folds. * Best-performing model. Bold denotes highest value per metric. SVM = Support Vector Machine; RBF = Radial Basis Function.*

4.6 SHAP Feature Importance Analysis

SHAP values were computed for the Random Forest model to quantify the contribution of each predictor to burnout classification decisions. Mean absolute SHAP values are presented in Table 6. Psychological distress was the most influential predictor of burnout classification (Mean |SHAP| = 0.231), demonstrating more than four times the predictive contribution of academic stress (Mean |SHAP| = 0.056). Social support ranked second (0.163) and self-efficacy third (0.110), with both exerting protective influences: higher social support and self-efficacy values were associated with reduced burnout probability (negative SHAP contributions), while lower values were associated with increased burnout

probability. The SHAP summary plot (Figure 5) confirmed that the direction of all effects was consistent with theoretical expectations and with the regression findings reported in Section 4.4.

Table 6

SHAP Feature Importance — Random Forest Classifier (N = 321)

Rank	Feature	Mean SHAP	Interpretation
1	Psychological Distress	0.231	Strongest predictor; high distress increases burnout probability
2	Social Support	0.163	Protective; higher support reduces burnout probability
3	Self-Efficacy	0.110	Protective; higher efficacy reduces burnout probability
4	Academic Stress	0.056	Weakest unique contribution when all predictors are modelled

Note. Mean |SHAP| = mean absolute SHAP value across all 321 observations, reflecting overall feature importance in the Random Forest model.

All five figures referenced throughout (Correlation Heatmap, Feature Importance Chart, ROC Curves, Burnout Distribution, and SHAP Summary Plot) are appended at the end of the document as Figures 1–5.

DISCUSSION

The present study sought to develop and evaluate machine learning models for predicting academic burnout among secondary school and undergraduate students, while employing SHAP analysis to illuminate the relative contributions of risk and protective psychological factors. Findings replicate and extend several established patterns in the burnout literature while also revealing important nuances regarding the differential contributions of academic stress and broader psychological distress as predictors.

5.1 The Primacy of Psychological Distress

The most consequential finding of the present study is the identification of psychological distress as the strongest predictor of burnout — both in the linear regression model ($B = 0.356$, $p < .001$) and in SHAP feature importance analysis (Mean $|\text{SHAP}| = 0.231$), where it contributed more than four times the predictive weight of academic stress. This finding is consistent with Maslach et al.'s [6] theoretical account of burnout as a product of progressive emotional resource depletion: students who experience chronic difficulties with emotional regulation, persistent worry, and coping under pressure may exhaust the psychological reserves necessary for sustained academic engagement more rapidly and severely than those experiencing domain-specific academic stress alone.

This result also aligns meaningfully with Gao et al.'s [3] moderated mediation model, which demonstrated that academic anxiety functions as a mediating mechanism through which stress contributes to burnout. The present findings suggest that psychological distress — which encompasses anxiety alongside broader emotional strain — may represent a more proximal and comprehensive predictor of burnout than academic stress per se. This distinction has important theoretical implications: it suggests that models of burnout development should incorporate broader emotional burden as a construct distinct from, and potentially more predictive than, domain-specific academic stress. For Walburg's [2] characterisation of academic pressure as a primary driver of burnout, the present data partially diverge — while stress is meaningfully correlated with burnout, its unique predictive contribution diminishes

substantially when psychological distress is simultaneously modelled, suggesting that stress may operate largely through the pathway of emotional distress rather than independently.

5.2 Protective Factors: Social Support and Self-Efficacy

Social support and self-efficacy both emerged as significant and theoretically meaningful protective factors. Social support demonstrated the second-largest SHAP contribution (0.163) and a significant independent regression coefficient ($B = -0.314$, $p < .001$), consistent with the broader literature on social support as a buffer against the adverse psychological effects of chronic stress. For students navigating the pressures of competitive secondary education or the demands of early university life in India, the perception that family support is available and that help-seeking pathways exist may provide a critical psychological resource — one that sustains engagement and prevents the progressive depletion underlying burnout.

Self-efficacy similarly demonstrated a significant independent contribution to burnout prediction ($B = -0.308$, $p < .001$; Mean $|\text{SHAP}| = 0.110$), directly replicating Gao et al.'s [3] finding that students with stronger academic self-efficacy are less susceptible to burnout under conditions of elevated stress. This finding is also consistent with Bandura's [8] theoretical prediction that self-efficacy beliefs buffer individuals against the resource depletion and learned helplessness processes that characterise burnout development. Higher self-efficacy may protect against burnout by enabling students to appraise academic demands as manageable challenges rather than overwhelming threats — thereby reducing the emotional strain that, when sustained, produces exhaustion and cynicism.

Taken together, the consistent identification of social support and self-efficacy as protective factors alongside distress and stress as risk factors supports a risk-and-resilience model of burnout in which vulnerability is determined not only by the presence of environmental and psychological stressors but by the availability of psychological and social protective resources. This framework carries direct implications for intervention design, as discussed in the Conclusion.

5.3 The Attenuated Role of Academic Stress

While academic stress demonstrated a significant bivariate association with burnout ($r = .441$, $p < .001$), it was not a significant independent predictor in the regression model ($B = -0.096$, $p = .107$) and showed the weakest SHAP contribution (0.056) in the machine learning analysis. This attenuation is most parsimoniously explained by the high intercorrelation between stress and distress ($r = .683$): when both are included in the same model, distress appears to subsume the variance in burnout previously attributable to stress when modelled alone. This pattern is consistent with Gao et al.'s [3] mediation framework, in which anxiety — a core component of the distress measure employed here — mediates the relationship between academic stress and burnout. The present results suggest that distress may represent a more distal consequence of academic stress that then more proximally drives burnout, rendering stress's direct contribution to burnout non-significant in the presence of distress.

5.4 Machine Learning Performance and Practical Implications

All four machine learning models demonstrated strong classification performance, with Random Forest achieving near-perfect discrimination (ROC-AUC = .998) and XGBoost performing comparably (ROC-AUC = .994). The strong performance of even the simplest model — Logistic Regression (ROC-AUC = .952, Accuracy = .879) — is substantively informative, indicating that the burnout signal is sufficiently strong and structurally consistent to be captured by linear classifiers, and that the predictive relationships identified are robust rather than dependent on complex nonlinear interactions. This convergence of evidence across models of varying complexity strengthens confidence in the reliability of the identified predictor–outcome relationships.

The consistently high performance metrics across all models raise the possibility of a scalable, low-cost screening application: a brief survey assessing the four key predictors could provide counsellors, educators, or school psychologists with a probabilistic burnout risk indicator, enabling earlier and more targeted intervention. However, as discussed in Section 5.5, significant methodological limitations — particularly the small sample size, cross-sectional design, and median-split binarisation — must be addressed before any clinical or institutional application could responsibly be considered. The present findings should be treated as preliminary and proof-of-concept.

5.5 Limitations

The findings of the present study must be interpreted in light of several important methodological limitations that constrain the strength of conclusions that can be drawn.

First, the cross-sectional research design precludes causal inference regarding the relationships between predictors and burnout. All variables were measured at a single point in time, making it impossible to determine whether psychological distress, low social support, or low self-efficacy precede and causally contribute to burnout development, or whether burnout itself produces elevated distress, diminished help-seeking, and reduced efficacy perceptions through feedback processes. Longitudinal studies employing repeated assessments across the academic year would be necessary to establish temporal precedence and support causal claims.

Second, the study relied exclusively on self-report measures, which are subject to well-documented limitations including social desirability bias, retrospective recall distortions, and response inconsistency across time points. Students may systematically underreport burnout or distress symptoms due to concerns regarding academic or social consequences, or conversely may overreport difficulties in response to current affective states. Future research would benefit from triangulating self-report data with objective indicators such as academic performance records, teacher ratings, absenteeism data, or physiological markers of stress.

Third, the sample was drawn from an Indian secondary school and early undergraduate context, limiting the generalisability of findings to other educational systems, cultural settings, and age groups. Educational environments vary substantially across countries in terms of academic pressure, institutional support infrastructure, and cultural norms regarding achievement, help-seeking, and psychological disclosure. The patterns observed in the present Indian sample may not replicate in Western, Scandinavian, or lower-pressure educational contexts. Replication across diverse national contexts is essential before broader generalisability can be claimed.

Fourth, the internal consistency of the Self-Efficacy ($\alpha = .265$) and Social Support ($\alpha = .572$) scales was low, primarily due to their two-item composition. With only two items, Cronbach's alpha is structurally constrained and provides an underestimate of true reliability independent of item quality.

Nevertheless, the limited measurement precision of these constructs may have attenuated their apparent predictive contributions. Future research should employ validated multi-item scales such as the Academic Self-Efficacy subscale from established instruments, and multi-item social support scales such as the Multidimensional Scale of Perceived Social Support [22].

Fifth, the Burnout composite demonstrated only questionable internal consistency ($\alpha = .655$), likely reflecting its multidimensional structure and the limited number of items per subscale. Future studies should employ the full MBI-SS or SBI instruments with their complete item pools to provide more reliable and dimensionally distinct burnout subscale scores, enabling separate analyses of exhaustion, cynicism, and academic efficacy dimensions.

Sixth, the machine learning performance metrics — while impressive — should be interpreted cautiously in light of the methodological choices that may inflate them. The binarisation of burnout at the sample median creates two relatively balanced and separable classes, and performance estimates derived from this approach may not generalise to scenarios involving population-representative burnout thresholds or continuous outcome prediction. Furthermore, a sample of $N = 321$ is modest for machine learning applications; cross-validation performance estimates in small samples may overfit to sample-specific patterns and may not replicate in larger, more demographically diverse datasets. External validation in independent samples is strongly recommended before any predictive claims are generalised.

Seventh, the high intercorrelation between academic stress and psychological distress ($r = .683$) raises concerns regarding multicollinearity in the regression model, which may have inflated standard errors and reduced the apparent unique contribution of academic stress. Variance inflation factor (VIF) diagnostics, ridge regression, or structural equation modelling approaches that explicitly model the shared variance between these constructs may be informative in future replications.

Eighth, sleep duration and screen time — both included in the survey and theoretically relevant to burnout — could not be incorporated into the primary machine learning models due to their categorical encoding. Future analyses should include these variables using appropriate numeric encoding or ordinal transformation, as they may contribute additional predictive variance and provide a more comprehensive picture of burnout risk factors.

CONCLUSION

The present study investigated the prediction of academic burnout among secondary school and undergraduate students in India using four machine learning classifiers trained on a theoretically grounded set of psychosocial predictors. Using five-fold stratified cross-validation with a sample of 321 students, the Random Forest classifier achieved the highest classification performance (Accuracy = 97.5%, ROC-AUC = .998), with XGBoost performing comparably. SHAP analysis identified psychological distress as the dominant predictor of burnout, followed by social support and self-efficacy as significant protective factors, with academic stress demonstrating the weakest independent contribution when all predictors were simultaneously modelled. Multiple linear regression corroborated these findings, with the four predictors collectively accounting for 58.3% of variance in burnout scores.

These findings carry several important implications for educational practice. First, they emphasise the critical importance of monitoring students' broader psychological well-being — not merely academic stress — as a primary indicator of burnout risk. Schools and institutions that focus exclusively on managing academic workload may overlook students whose burnout is driven primarily by broader emotional distress that extends beyond the academic domain. Holistic well-being screening programmes that assess psychological distress alongside academic pressure would provide a more sensitive and comprehensive approach to early identification.

Second, the strong protective effects of social support and self-efficacy identified in the present study suggest that interventions targeting burnout should not only aim to reduce academic stressors but should actively cultivate students' protective psychological resources. Initiatives that strengthen students' awareness of and access to support resources — including mentoring programmes, peer support networks, school counselling services, and family engagement strategies — may be as impactful as academic load management approaches. Psychological skill-building programmes designed to develop academic self-efficacy, adaptive coping strategies, and resilience may similarly offer significant preventive value, particularly for students navigating the high-pressure transitions of senior secondary school and early undergraduate study.

Third, from a methodological standpoint, the study demonstrates the feasibility of applying machine learning to burnout screening using a small number of psychometrically grounded survey items. Even the simplest classifier achieved 87.9% accuracy, suggesting that brief, low-cost assessment instruments could, in principle, provide educationally meaningful risk indicators. The integration of SHAP analysis addresses the interpretability concerns that have limited adoption of machine learning in educational settings, demonstrating that complex predictive models need not sacrifice transparency.

Future research should address the present study's limitations through longitudinal designs, validated multi-item measurement instruments, larger and more demographically diverse samples spanning multiple national contexts, and continuous burnout outcome modelling. The inclusion of additional theoretically relevant predictors — including sleep quality, academic performance, peer relationships, and screen time — within the machine learning feature set would further enrich the predictive framework. Extending the present approach to examine burnout trajectories over time, and to evaluate whether machine learning-informed early interventions produce meaningful reductions in burnout incidence, represents an important and practically valuable direction for future research.

In conclusion, this study provides preliminary but substantive evidence that explainable machine learning offers a promising and practically viable framework for the early identification of students at risk of academic burnout. By integrating theoretically grounded psychological constructs with contemporary computational methods, and by prioritising model interpretability alongside predictive performance, the present research contributes to both the scientific understanding of burnout mechanisms and the institutional capacity for proactive, data-informed student support. The consistent identification of psychological distress, social support, and self-efficacy as the dominant drivers of burnout risk across both statistical and machine learning analyses provides a coherent and actionable evidence base for the design of targeted preventive interventions in secondary school and undergraduate educational settings.

CONFLICT OF INTEREST

The author declares that there are no conflicts of interest related to this work.

REFERENCES

1. Maslach C, Jackson SE. The measurement of experienced burnout. *J Organ Behav*. 1981;2(2):99–113.
2. Walburg V. Burnout among high school students: a literature review. *Child Youth Serv Rev*. 2014;42:28–33.
3. Gao W, Ping S, Liu X. Academic stress and burnout among adolescents: the moderated mediation role of academic anxiety and self-efficacy. *Int J Environ Res Public Health*. 2023;20(3):2019.
4. Bask M, Salmela-Aro K. Burned out to drop out: exploring the relationship between school burnout and school dropout. *Eur J Psychol Educ*. 2013;28(2):511–528.
5. Schaufeli WB, Martínez IM, Pinto AM, Salanova M, Bakker AB. Burnout and engagement in university students: a cross-national study. *J Cross Cult Psychol*. 2002;33(5):464–481.
6. Maslach C, Schaufeli WB, Leiter MP. Job burnout. *Annu Rev Psychol*. 2001;52(1):397–422.
7. Bandura A. Self-efficacy: toward a unifying theory of behavioral change. *Psychol Rev*. 1977;84(2):191–215.
8. Bandura A. *Self-Efficacy: The Exercise of Control*. New York: W. H. Freeman; 1997.
9. Romero C, Ventura S. Educational data mining and learning analytics: an updated survey. *WIREs Data Min Knowl Discov*. 2020;10(3):e1355.
10. Shatte ABR, Hutchinson DM, Teague SJ. Machine learning in mental health: a scoping review of methods and applications. *Psychol Med*. 2019;49(9):1426–1448.
11. Lundberg SM, Lee SI. A unified approach to interpreting model predictions. *Adv Neural Inf Process Syst*. 2017;30:4765–4774.
12. Pekrun R. The control-value theory of achievement emotions: assumptions, corollaries, and implications for educational research and practice. *Educ Psychol Rev*. 2006;18(4):315–341.
13. Salmela-Aro K, Kiuru N, Leskinen E, Nurmi JE. School Burnout Inventory (SBI): reliability and validity. *Eur J Psychol Assess*. 2009;25(1):48–57.
14. Graham S, Depp C, Lee EE, Nebeker C, Tu X, Kim HC, Jeste DV. Artificial intelligence for mental health and mental illnesses: an overview. *Curr Psychiatry Rep*. 2019;21(11):116.
15. Baker RS, Inventado PS. Educational data mining and learning analytics. In: Larusson JA, White B, editors. *Learning Analytics: From Research to Practice*. New York: Springer; 2014. p. 61–75.
16. Harrer M, Adam SH, Baumeister H, Cuijpers P, Karyotaki E, Auerbach RP, Kessler RC, Bruffaerts R, Berking M, Ebert DD. Internet interventions for mental health in university students: a systematic review and meta-analysis. *Int J Methods Psychiatr Res*. 2021;30(2):e1878.
17. Ribeiro MT, Singh S, Guestrin C. “Why should I trust you?”: explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*; 2016. p. 1135–1144.
18. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: machine learning in Python. *J Mach Learn Res*. 2011;12:2825–2830.
19. Chen T, Guestrin C. XGBoost: a scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*; 2016. p. 785–794.
20. Seabold S, Perktold J. Statsmodels: econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*; 2010. p. 57–61.
21. Vallat R. Pingouin: statistics in Python. *J Open Source Softw*. 2018;3(31):1026.

22. Zimet GD, Dahlem NW, Zimet SG, Farley GK. The multidimensional scale of perceived social support. *J Pers Assess.* 1988;52(1):30–41.

Appendix A

Scale Descriptions and Item Content

The following section describes each composite scale employed in the present study, including the full wording of constituent items, response format, scoring procedure, and internal consistency estimates. All items were administered as part of a single structured online survey via Google Forms.

A.1 Academic Stress Scale (5 Items)

Response format: Five-point Likert-type scale (1 = Not at all / Strongly Disagree; 5 = Very Often / Strongly Agree). Scoring: unweighted mean of all five items; higher scores indicate greater academic stress. Cronbach's $\alpha = .823$.

Table A.1

Academic Stress Scale — Item Wording

Item	Wording
AS1	How often do you feel overwhelmed by schoolwork?
AS2	I feel pressure to achieve high grades.
AS3	I worry about my academic performance.
AS4	I often worry about upcoming tests or examinations.
AS5	I struggle to balance schoolwork with other responsibilities.

A.2 Psychological Distress Scale (3 Items)

Response format: Five-point Likert-type scale (1 = Never; 5 = Very Often), referring to the preceding month. Scoring: unweighted mean of all three items; higher scores indicate greater distress. Cronbach's $\alpha = .827$.

Table A.2*Psychological Distress Scale — Item Wording*

Item	Wording
PD1	Over the last month, how often have you felt emotionally overwhelmed?
PD2	Over the last month, how often have you struggled to cope with stress?
PD3	Over the last month, how often have you felt persistently worried?

A.3 Academic Burnout Composite (7 Items)

Response format: Five-point Likert-type scale (1 = Never / Strongly Disagree; 5 = Always / Strongly Agree). Scoring: Items BE1–BE3 (Exhaustion) and BC1–BC2 (Cynicism) scored directly; items BAE1–BAE2 (Academic Efficacy) reverse-scored (6 minus raw score) prior to inclusion. Composite burnout score = unweighted mean of all seven items (including reverse-coded efficacy items); higher scores indicate greater burnout. Cronbach's $\alpha = .655$.

Table A.3*Academic Burnout Composite — Item Wording and Subscale Assignment*

Item	Subscale	Wording
BE1	Exhaustion	I feel exhausted at the end of a typical school day.
BE2	Exhaustion	I feel emotionally drained by my studies.
BE3	Exhaustion	I often feel too tired to complete academic tasks.
BC1	Cynicism	I have become less interested in my studies over time.
BC2	Cynicism	I sometimes question whether my schoolwork is meaningful.
BAE1	Efficacy (R)	I am confident in my ability to handle difficult school assignments. [Reverse-scored]
BAE2	Efficacy (R)	I believe I can achieve my academic goals. [Reverse-scored]

A.4 Self-Efficacy Scale (2 Items)

Response format: Five-point Likert-type scale (1 = Strongly Disagree; 5 = Strongly Agree). Scoring: unweighted mean of both items; higher scores indicate greater self-efficacy. Cronbach's $\alpha = .265$ (structurally limited by two-item composition; inter-item $r = .26$).

Table A.4

Self-Efficacy Scale — Item Wording

Item	Wording
SE1	I am confident in my ability to handle difficult school assignments.
SE2	I believe I can achieve my academic goals.

A.5 Social Support Scale (2 Items)

Response format: Five-point Likert-type scale (1 = Strongly Disagree; 5 = Strongly Agree). Scoring: unweighted mean of both items; higher scores indicate greater perceived social support. Cronbach's $\alpha = .572$ (structurally limited by two-item composition).

Table A.5

Social Support Scale — Item Wording

Item	Wording
SS1	My family supports me during difficult academic periods.
SS2	I know where to seek help when I face challenges.

Note. Figures 1–5 (Correlation Heatmap, Feature Importance Chart, ROC Curves, Burnout Distribution, and SHAP Summary Plot) are provided as separate image files to be inserted at the end of the document or within Section 4 as appropriate for the target journal's formatting requirements.